

MODULE 03

Statistics for Data Science

AI with Data Science - Professional Learning Series

ARSHITH LMS | Professional Edition

Statistics for Data Science

Descriptive Statistics

Descriptive statistics summarize and describe data.

Measures of Central Tendency:

Mean: average value

Median: middle value (robust to outliers)

Mode: most frequent value

Measures of Spread:

Variance: average squared deviation from mean

Std Deviation: square root of variance

Range: max - min

IQR: Q3 - Q1 (interquartile range)

Probability Basics

Probability = favorable outcomes / total outcomes

Key Rules:

Addition: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

Multiplication: $P(A \text{ and } B) = P(A) \times P(B|A)$

Complement: $P(\text{not } A) = 1 - P(A)$

Bayes Theorem: $P(A|B) = P(B|A) \times P(A) / P(B)$

Probability Distributions

Normal Distribution: bell curve, mean=0, std=1

Binomial: discrete, n trials, p probability

Poisson: count of events in time interval

Uniform: equal probability for all outcomes

```
import scipy.stats as stats
# Normal distribution
stats.norm.pdf(x, mean, std)
stats.norm.cdf(x, mean, std)
```

Hypothesis Testing

1. State H_0 (null) and H_1 (alternative)
2. Choose significance level ($\alpha = 0.05$)
3. Collect data and compute test statistic

4. Calculate p-value
5. Reject H_0 if $p\text{-value} < \alpha$

Common Tests:

t-test: compare means

chi-square: categorical independence

ANOVA: compare 3+ groups

Correlation and Regression

Correlation: measures linear relationship (-1 to 1)

> 0.7: strong positive

< -0.7: strong negative

~ 0: no relationship

Pearson correlation

`df.corr()`

Simple linear regression

$y = b_0 + b_1x + \text{error}$

R-squared: proportion of variance explained